A Bayesian model of color constancy using a physics-based generative process

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Abstract

The same object can reflect a different spectrum to the retina under different illuminations, yet human typically maintain a stable perception of object color, an ability known as color constancy. We present a Bayesian computational model of color constancy that investigates the role of surface specularity in resolving the ambiguity between object reflectance and illumination. Drawing on recent psychophysical evidence suggesting that specular highlights facilitate scene discriminability, we model the image formation process using a physically accurate renderer Mitsuba to infer latent scene parametersspecifically object chromaticity, illumination chromaticity, and glossiness. To estimate the posterior distribution effectively, we employ a hybrid inference strategy that combines an amortized multi-layer perceptron with a Random Walk Metropolis-Hastings algorithm. We evaluated the model across five distinct illumination and object chromaticities in the CIE $u'v'$ space, comparing performance between matte and glossy surfaces. Our results demonstrate a functional dissociation in the utility of specular cues: the presence of glossiness marginally improves object color constancy (p=0.0945) but significantly impairs illumination estimation (p=0.041). Furthermore, an analysis of error distributions reveals chromaticity-dependent asymmetries that parallel human psychophysical benchmarks. These findings suggest that while specular cues constrain surface reflectance properties, they may introduce confounding variance in illuminant estimation, offering new insights into the computational trade-offs required for visual stability.

1. Introduction

1.1. Human Color Constancy

Color constancy refers to the visual system's capacity to maintain a stable perception of object surface colors despite significant variations in the spectral composition of the illumination [12]. This perceptual stability is computationally non-trivial because the light signal reaching the retina is a product of two confounding factors: the surface reflectance of the object and the spectral power distribution of the illuminant [2]. Mathematically, this creates an ill-posed inverse problem where an infinite number of reflectance and illumination combinations can result in the same observed pixel values.

Despite this ambiguity, human observers demonstrate remarkable constancy in naturalistic environments. Psychophysical research suggests that the visual system relies on statistical regularities in natural scenessuch as the gray world hypothesis [3] or the tendency of natural illuminants to lie near the Planckian locus [4]to constrain this problem. However, global scene statistics alone often fail to explain constancy in complex 3D environments, suggesting that the visual system must also exploit local cues derived from material properties and scene geometry [7].

1.2. The Role of Specularity and Material Properties

While early models of color constancy treated objects as purely Lambertian (matte) surfaces, recent evidence highlights the critical role of material propertiesspecifically glossiness and specularityin scene disambiguation. Specular highlights act as clues rather than noise; because they reflect the illuminant spectrum more directly than diffuse reflections, they potentially offer a strong signal for recovering the illumination chromaticity [13].

Recent psychophysical studies support this specularity-as-signal hypothesis. For instance, Virtanen et al. (2025) demonstrated that the presence of specular highlights in 3D scenes significantly enhances the discriminability of reflectance changes compared to matte scenes, suggesting that glossiness aids in separating surface color from lighting [10]. Similarly, experimental work by Lee and Smithson (2016) indicates that observers can utilize specular highlights to maintain operational color constancy even when other contextual cues are weakened [9]. However, the inte-

Code is available at: https://github.com/Jinniesss/Color-Constancy



Figure 1. Color constancy. Two photographs of a scene taken under a reddish and greenish illumination. The bar illustrates that the colorimetric color of the grape in the left photograph under red illumination is the same as the colorimetric color of the orange in the right photograph under green illumination. Because of color constancy, we identify the surface color of the grapes in the left photograph with the surface color of the grapes in the right photograph, not with the color of the orange in the right photograph. Nevertheless, the color of the grapes does not look the same in both photographs, implying that some level of interference is involved in identifying the colors across the two photographs. Figure reproduced from [12].

gration of these cues is not always beneficial; highly specular surfaces can introduce high-variance signals that may confound estimation under specific viewing conditions [11].

1.3. Computational Approaches to Inverse Rendering

In computer vision, the problem of disentangling these latent scene parameters is often framed as inverse rendering. Traditional approaches have relied on handcrafted heuristics or statistical assumptions (e.g., Retinex theory, White Patch algorithm) that often break down in non-standard lighting conditions [5]. More recently, data-driven approaches using deep neural networks have achieved high performance but often lack interpretability and fail to generalize outside their training distribution.

A promising alternative is analysis by synthesis, or physics-based generative modeling. By explicitly modeling the physical transport of lightincluding diffuse and specular interreflectionsthese models can "invert" the generative process to infer the most likely scene parameters that explain the observed image [14] [8]. This Bayesian formulation allows for the rigorous incorporation of prior knowledge about natural illumination and material statistics, providing a robust framework for investigating the computational mechanisms of constancy [1].

1.4. Present Work and Contributions

In this paper, we present a Bayesian computational model of color constancy that explicitly investigates the trade-offs introduced by surface specularity. We model the image formation process using a physically accurate renderer Mitsuba [6] to infer latent scene parametersspecifically object chromaticity, illumination chromaticity, and glossiness.

Our approach utilizes a hybrid inference strategy, combining an amortized Multi-Layer Perceptron (MLP) with a Random Walk Metropolis-Hastings algorithm to estimate the posterior distribution of scene properties. We evaluate the model's performance across diverse illumination and object colors in the CIE $u'v'$ space, comparing results between matte and glossy conditions. Our findings reveal a functional dissociation: while glossiness marginally improves the recovery of object surface color, it introduces significant variance in illumination estimation. Furthermore, we observe distinct asymmetries in error rates across illumination chromaticities (e.g., Blue vs. Red light) that parallel human psychophysical biases documented in literature [4].

2. Methods

2.1. Generative Model

We define a generative probabilistic model that links latent scene parameters to observed image data through physical light transport. The image formation process is modeled using the Mitsuba 3 physically-based renderer (configured in RGB mode) [6]. We define the rendering process as a deterministic function f_ψ that maps scene parameters to a rendered image $\hat{y}$.

Our latent vector contains both chromaticity coordinates (defined in the perceptually uniform CIE $u'v'$ space) and material properties. Since the renderer operates in linear RGB space, we introduce a color transformation function f_γ specifically for the chromatic components. The glossiness parameter does not require transformation and is passed directly. The generative process is thus defined as:

$$\hat{y} = f_\psi(f_\gamma(\theta_c), \alpha, S) \tag{1}$$

where S represents the fixed scene geometry and repre-

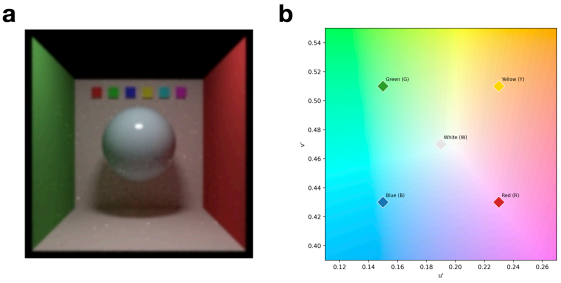


Figure 2. **a** The rendered 3D scene configuration, featuring a central target sphere under directional illumination. The back wall contains a "Mondrian" pattern of matte colored patches to provide essential local chromatic context. **b** The sampling grid in the CIE $u'v'$ color space used for both object and illumination chromaticities. The five test points (Red, Green, Blue, Yellow, White) form a symmetrical cross pattern centered on the neutral white point.

sents the latent variables; $_c = \{u_O, v_O, u_L, v_L\}$ represents the object and illumination chromaticities; f_γ converts these uv coordinates into the valid RGB reflectance and emission spectra required by Mitsuba; α is the scalar glossiness parameter. For the roughness parameter, lower values (e.g., 0.01) correspond to sharp specular highlights (glossy), while higher values (e.g., 0.99) correspond to a purely diffuse appearance (matte).

Scene Configuration (S): The fixed scene S models a controlled 3D environment designed to simulate a standard psychophysical setup. The scene consists of a central spherical target object positioned in the middle of a room. To provide chromatic context cues essential for color constancy (i.e., local contrast), the back wall features a spatial arrangement of distinct colored patches, creating a Mondrian-like background. The illumination is provided by a directional light source positioned high and forward relative to the object, ensuring clear visibility of surface shading and specular highlights.

Latent Variables (θ): The full vector θ represents the learnable latent variables we seek to infer:

$$\theta = \{u'_O, v'_O, u'_L, v'_L, \alpha\} \tag{2}$$

where (u_O, v_O) denote the object chromaticity of the central sphere; (u_L, v_L) denote the illumination chromaticity of the light source; α represents the object glossiness (specular roughness) of the central sphere.

The forward function f_ψ simulates complex light transport phenomena, including diffuse interreflections between the wall patches and the object, as well as specular highlights on the target sphere. We formulate the inversion as a Bayesian inference task, estimating the posterior distribution given an observed image y_{obs}:

$$P(u'_O, v'_O, u'_L, v'_L, \alpha | y_{obs}) \propto P(y_{obs} | f_\psi(\theta, S)) \cdot P(\theta) \tag{3}$$

We assume a uniform prior $P(\theta)$ over the valid parameter space bounded by the CIE $u'v'$ gamut and the physical limits of roughness. Specifically, the priors are defined over the ranges $u' \in [0.11, 0.27]$, $v' \in [0.39, 0.55]$, and $\alpha \in [0.01, 0.99]$. The likelihood function is modeled as a Gaussian distribution centered on the rendered output:

$$P(y_{obs}|\theta) \propto \exp\left(-\frac{||y_{obs} - f_\psi(f_\gamma(\theta_c), \alpha, S)||_2^2}{2\sigma^2}\right) \tag{4}$$

2.2. Hybrid Inference Strategy

Estimating the posterior distribution directly via sampling is computationally prohibitive due to the high cost of evaluating the rendering function for every proposal. To overcome this bottleneck, we employ a hybrid inference strategy that combines the speed of amortized neural inference with the accuracy of Markov Chain Monte Carlo (MCMC) refinement.

Amortized Initialization: To initialize the inference chain in a high-probability region of the posterior, we first employ a neural network regressor. We trained a Multi-Layer Perceptron (MLP) to approximate the inverse mapping from the image space back to the parameter space. The network was trained on a synthetic dataset of 1000 latent parameter-image pairs generated from the prior. Given an observed test image, the MLP outputs a point estimate for the scene parameters. This estimate serves as the starting state for the sampling process, effectively eliminating the need for a lengthy burn-in period.

MCMC Refinement: Starting from the neural initialization, we employ the Random Walk Metropolis-Hastings (RWMH) algorithm to explore the local posterior geometry and refine the estimate. At each step, a candidate state is proposed using a truncated normal distribution centered on the current state, ensuring all parameters remain within valid physical and colorimetric bounds. The proposal distributions are parameterized by two distinct standard deviations: $\sigma_c = 0.04$ for both object and illumination chromaticity coordinates, and $\sigma_{gloss} = 0.25$ for the glossiness parameter. The candidate is accepted or rejected based on the Metropolis ratio, which compares the likelihood of the candidate against the current state by re-rendering the scene. This refinement step allows the model to correct for approximation errors in the neural network and to quantify the uncertainty of the estimate, which is critical for analyzing the trade-offs between glossy and matte conditions.

2.3. Stimuli and Experimental Design

To evaluate the model, we generated a comprehensive test set of synthetic images featuring a central 3D target object illuminated by a directional light source. We employed a full factorial design crossing five object chromaticities,

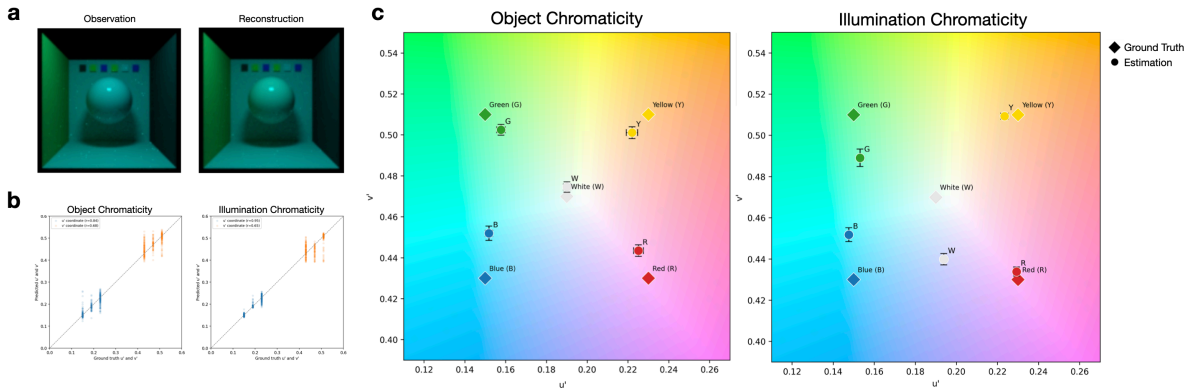


Figure 3. Model performance. **a** A representative input observation (y_{obs}) compared with the reconstructed image ($\hat{y}$) rendered using the inferred posterior mean parameters. The close visual similarity demonstrates the model's ability to accurately explain the observed pixel data. **b** Scatter plots comparing the estimated posterior mean chromaticities against ground truth values for Object (left) and Illumination (right) across all 400 trials. The dashed diagonal identity line represents perfect recovery. **c** The mean posterior estimates (circles) are plotted against ground truth chromaticities (diamonds) in the CIE $u'v'$ space for Object (left) and Illumination (right) recovery.

five illumination chromaticities, and two surface material conditions. This resulted in 50 unique experimental settings. To ensure robust statistical estimates and account for the stochastic nature of the MCMC sampling process, we executed 8 independent trials for each setting, yielding a total of 400 experimental runs.

The specific chromaticity values were selected to form a symmetrical cross pattern in the CIE $u'v'$ color space, ensuring balanced sampling along the cardinal axes relative to a neutral white point. The five chromaticity conditions were defined as follows: Neutral White ($u' = 0.19$, $v' = 0.47$), Red ($u' = 0.23$, $v' = 0.43$), Green ($u' = 0.15$, $v' = 0.51$), Blue ($u' = 0.15$, $v' = 0.43$), and Yellow ($u' = 0.23$, $v' = 0.51$).

For the material conditions, we manipulated the roughness parameter to simulate distinct surface properties while holding the chromaticity constant. The matte condition was modeled using a high roughness value ($\alpha = 0.99$) to approximate a pure Lambertian diffuse reflectance. Conversely, the glossy condition was modeled using a low roughness value ($\alpha = 0.01$) within a Ward BRDF model, producing sharp, distinct specular highlights on the object surface. For each trial, the model's performance was quantified by calculating the Euclidean distance between the estimated posterior mean and the ground truth chromaticity coordinates for both the object and the illumination.

3. Results

3.1. Overall Model Performance

We first validated the model's ability to recover latent scene parameters across all experimental trials. Figure 3 illustrates both the reconstruction quality and estimation accuracy.

As shown in Figure 3a, the image rendered using the inferred posterior mean closely matches the input observation (y_{obs}), confirming that the model converged to a solution that accurately explains the sensory data.

Figure 3b details the accuracy of parameter recovery. The posterior mean estimates for both object (left) and illumination (right) chromaticities cluster tightly along the $y = x$ identity line, demonstrating robust color constancy. Notably, object color estimates form distinct, well-separated clusters for each chromatic category, indicating that the model successfully discounts the illuminant. Quantitatively, the model achieves high linear correlation with ground truth values, though performance varies by chromatic axis. For object recovery, the model performs better on the red-green axis ($u' : r = 0.84$) compared to the yellow-blue axis ($v' : r = 0.68$). A similar trend is observed in illumination recovery ($u' : r = 0.95$ vs. $v' : r = 0.65$). Figure 3c further visualizes these estimates in the CIE $u'v'$ space.

3.2. The Effect of Glossiness on Constancy

To test the hypothesis that specular highlights facilitate scene disambiguation, we analyzed the specific impact of surface material properties on model performance. We calculated the differential error ($E_{glossy} - E_{matte}$) for each matched experimental condition, isolating the contribution of the specular component while holding geometry and chromaticity constant.

Figure 4 presents the distribution of these error differences. Our analysis reveals a functional dissociation in the utility of specular cues. For object chromaticity estimation, the presence of glossiness resulted in a marginal reduction in error ($p = 0.0945$). This suggests that specular highlights may provide a weak constraining signal that helps the inference algorithm anchor the surface reflectance, potentially

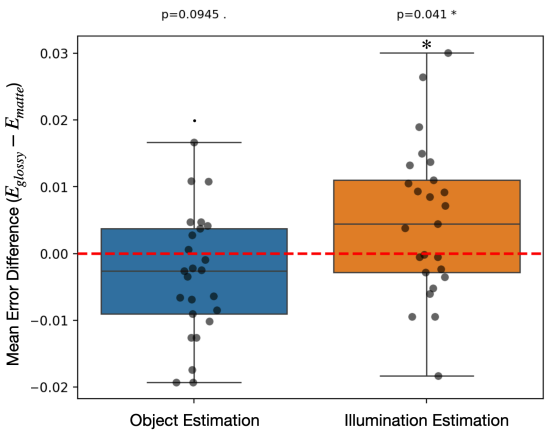


Figure 4. Effect of surface glossiness on estimation accuracy. Boxplots illustrating the distribution of error differences between glossy ($\alpha = 0.01$) and matte ($\alpha = 0.99$) conditions across all chromaticity settings. The y-axis represents the difference in Euclidean error ($E_{glossy} - E_{matte}$). Significance markers indicate: $^{\cdot}p < 0.1, ^{*}p < 0.05$.

by differentiating the diffuse body color from the interface reflection.

Conversely, for illumination chromaticity estimation, the addition of glossiness significantly impaired performance ($p = 0.041$). While theoretically, a specular highlight contains a direct reflection of the illuminant spectrum, our results indicate that this cue may introduce confounding variance rather than clarity. The high dynamic range and spatial sparsity of specular highlights likely create a more complex posterior landscape, making it more difficult for the model to converge on a stable estimate of the global illuminant compared to the fully diffuse signal available in the matte condition. This finding challenges the assumption that specular highlights are universally beneficial, suggesting instead that they introduce a trade-off between surface recoverability and illuminant stability.

3.3. Illuminant-Dependent Performance and Comparative Analysis

Our evaluation reveals that the model's color constancy performance is not uniform across the chromaticity of the illuminant. We observed a significant chromaticity-dependent bias in the inference error. Specifically, the model achieves its highest accuracy (lowest error) under Red illumination, while exhibiting its poorest performance (highest error) under Blue illumination.

This pattern of performance stands in direct contrast to psychophysical measurements of human color constancy, specifically those reported by Delahunt and Brainard [4]. In their experiment, human observers demonstrated the lowest constancy indices for Red illuminant changes. Conversely,

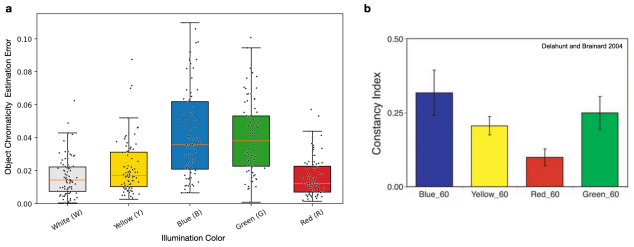


Figure 5. Comparison of generative model inference error versus human color constancy. **a** Mean chromaticity estimation error of the proposed model under four distinct illuminant conditions. Note that *lower*values indicate better performance. The model exhibits a bias toward long wavelengths, with minimum error under Red illumination and maximum error under Blue. **b** Human Constancy Indices (CI) from Delahunt and Brainard (2004) [4]. Here, higher values indicate superior constancy.

human performance was significantly more robust for Blue and Yellow illuminants, which align with the statistical regularities of natural daylight. Our model, therefore, exhibits an anti-daylight bias, excelling in long-wavelength dominance (Red) where human vision struggles, and failing in short-wavelength dominance (Blue) where human vision is typically robust.

Delahunt and Brainard calculated the Constancy Index (CI) using an equivalent illuminant method. Observers adjusted a test patch until it appeared achromatic (neutral gray) under a test illuminant. The CI was then computed as $CI = 1 - \frac{|e_2 - e_{eq}|}{|e_2 - e_1|}$, where e_1 is the chromaticity vector of the reference illuminant (neutral D65), e_2 is the test illuminant, and e_{eq} is the equivalent illuminant – the chromaticity that, if present in the reference scene, would produce the same visual response as the observers setting. A CI of 1 indicates perfect constancy (the observer perfectly accounted for the illuminant shift), while a CI of 0 indicates no constancy. Our model's error metric is inversely related to this index: high model error corresponds to a low human CI.

It is critical to interpret this divergence in the context of the differing experimental paradigms. The human data is derived from a perceptual achromatic adjustment task, relying on subjective appearance and potentially influenced by evolutionary priors for daylight distributions. In contrast, our generative model performs direct parametric inferenceover the color space to estimate the object's spectral or tristimulus values.

4. Discussion

4.1. Divergence in Specular Cues

A central motivation of this study was to test the hypothesis that specular highlights facilitate scene disambiguation. Recent psychophysical evidence indicates that for

human observers, the presence of glossiness improves discriminability for both surface reflectance and illumination changes [10]. Our model, however, exhibits a functional dissociation: while glossiness marginally improved object color recovery, it significantly impaired illumination estimation.

This divergence suggests a fundamental difference between the pixel-wise inference utilized by our model and the perceptual mechanisms of the human visual system. Humans likely rely on mid-level visual features – such as the geometric consistency of highlight placement or the separation of diffuse and specular layers – to infer illumination. These features are robust to small spatial variations. In contrast, our model relies on a strict pixel-wise likelihood function (L2 loss). Because specular highlights are high-frequency, spatially sparse signals, a minute misalignment in the estimated light position causes the rendered highlight to miss the target pixels entirely, resulting in a large error signal.

This finding implies that while specular highlights contain high-fidelity spectral information, they form a brittle cue for computational systems relying on direct synthesis. The human visual system's superior performance suggests it employs a more robust, perhaps summary statistic based representation of specularity that avoids the pitfalls of precise spatial alignment, a strategy that future computational models of constancy should emulate.

4.2. Divergence in Chromatic Biases

A comparison of our models error distribution with human psychophysics reveals a distinct anti-correlation. Our results indicate that the model performs robustly under Red and Yellow illumination but exhibits significant instability under Blue and Green illumination. In contrast, human observers typically demonstrate the opposite pattern. As reported by Delahunt and Brainard (2004), human color constancy is often most robust for illumination changes consistent with natural daylight (Blue) and degrades for atypical illuminants (Red) [4]. This divergence indicates a fundamental difference between the physics-based inference of our model and the ecologically-tuned inference of the human visual system.

The superior human performance under blue/daylight illumination strongly supports the daylight prior hypothesis – the idea that the visual system incorporates strong prior knowledge that natural illumination varies along a specific locus. Our model, conversely, employed a uniform prior, treating all illumination colors as equally probable. This suggests that the physics of light transport alone is insufficient to explain human competency; specific ecological priors are computationally necessary to resolve the high ambiguity of natural lighting conditions.

Furthermore, the model's degradation under Blue lights may be exacerbated by the non-isometric mapping between the inference space (CIE $u'v'$) and the rendering space (Linear RGB). The renderer's sensitivity to error – defined by the L2 pixel loss – is not uniform across the color gamut. It is likely that the likelihood landscape is flatter for variations in the blue region of the spectrum compared to red. This leads to wider posterior distributions and higher uncertainty, mirroring the difficulty humans face with uncommon lights where they lack the sharp prior to constrain these broad likelihoods. This emphasizes that robust constancy requires the alignment of both sensory precision and statistical expectations.

4.3. Limitations and Future Directions

Two primary limitations of the current implementation define the roadmap for future research.

A significant technical limitation identified in our analysis is the non-isometric mapping between the perceptual inference space (CIE $u'v'$) and the renderers linear RGB color space. This conversion inevitably introduces metric distortions, likely contributing to the artificially high uncertainty observed in the blue-yellow direction. To align the computational model more closely with physical reality, future iterations will transition from RGB-based rendering to full spectral rendering. By simulating light transport using wavelength-dependent spectral power distributions rather than trichromatic approximations, we can eliminate gamut-related artifacts and metamerism issues. This would allow us to isolate the information-theoretic limits of the inverse problem without the confounding influence of color space non-linearities.

Our results suggest that a uniform prior is insufficient to replicate human performance, particularly under natural daylight conditions. A key direction for future work is to systematically explore the prior space to identify the optimal probabilistic constraints for color constancy. Rather than manually imposing a daylight prior, we propose an ideal observer analysis: determining which prior distribution maximizes estimation accuracy given the physics of the environment. Furthermore, by fitting the parameters of the prior to match human psychophysical error patterns (e.g., the specific shape of the blue bias), we can effectively reverse-engineer the internal assumptions of the human visual system. This approach would allow us to quantitatively test the degree to which human color constancy is tuned to the statistical regularities of the natural world.

In summary, this study established a Bayesian inverse rendering framework to dissect the computational mechanisms of color constancy. Our results reveal a functional dissociation in material cues: while specular highlights aid in surface reflectance recovery, their spatial sparsity creates instability for illumination estimation. Furthermore, the models specific degradation under natural daylight con-

ditionsin direct contrast to robust human performanceunderscores the necessity of ecological priors to resolve physical ambiguities. Together, these findings suggest that achieving human-like constancy requires not only the integration of complex material cues but also strong top-down constraints tuned to the statistical regularities of the visual environment.

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